

4.4 Adding JavaScript and CSS to themes

As the internet evolves and more emphasis is placed on interactivity and user experience, the need for JavaScript increases. Using JavaScript has evolved from being painfully difficult, requiring browser detection and massive duplication of code to achieve a consistent user experience, to relatively simple, with implementations that work across all browsers. What once took developers 50 lines of code or more to achieve can now be done in just a few lines and is more reliable. JavaScript is a client-side language, meaning it is not executed on a server as PHP is. Instead, it relies on Web browsers to interpret it and handle it accordingly. The benefit is a simplistic concept of distributed computing, where the weight of processing and execution is not handled in a central spot, but distributed across all the computers accessing the Web page or Web application. In recent years, JavaScript has also been a cornerstone of highly interactive Web applications.

With the advent of Ajax (Asynchronous JavaScript and XML), Web applications gained the capability to use a hybrid model, where most of the user experience happens on the front end while background communication with servers handles most of the processing. This hybrid model provides a much more robust way to gain all the distribution and user experience of JavaScript with all the security and dynamic database communication of a server side system, such as PHP and MySQL. At the same time, user experience and themes require Cascading Style Sheets (CSS). CSS, like JavaScript, is a client-side technology. It is handled and parsed directly by the browser. There are a few versions of CSS, with varying levels of support from the various browsers and browser versions.

4.5 Examining the jQuery and Prototype Frameworks

WordPress makes it easy for developers to use the most common JavaScript frameworks available, jQuery and Prototype, and the Prototype-dependent framework, script.aculo.us. Most JavaScript in WordPress Admin is written in jQuery because it is a lightweight, but very powerful, library. However, script.aculo.us/Prototype provides richer animations and can be a more familiar development environment for some developers.

jQuery

jQuery, documented at www.jquery.com, powers most of the JavaScript in the WordPress Admin. However, you can also use it in themes by enqueueing (which means “adding a new item to the queue”) the script in a function or plugins.